

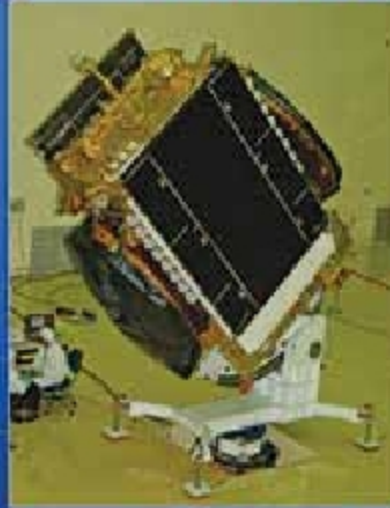
part of satellite testing.

Modern computer-controlled test equipment also plays a crucial role in modern satellite testing. Detailed knowledge of antennas, microwave, RF, IF, and baseband engineering is essentially required for satellite test engineers. Further, testing techniques of many laboratories are significantly modified by taking into account factors such as rapid spacecraft temperature changes, doppler shift, propagation fluctuations, and signal propagation delay.

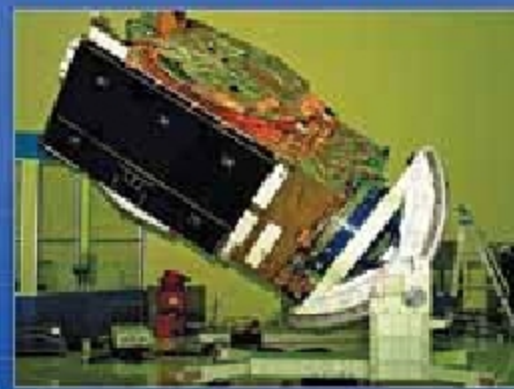
Small satellites stay at low earth or medium earth orbit and have life expectancies from several weeks to four years or more. The testing of such satellites helps both financially and physically. Space simulation testing is another part of satellite testing. In this, advanced chambers are used to imitate space temperature, pressure, and humidity.

A single satellite can have dozens to hundreds of spot beams. For example, the Inmarsat GEO I-4 satellite has 200 narrow spot beams. Spot beams allow satellites to transmit different data signals using the same frequency. Spot beams greatly increase the complexity of design, test, and calibration of antennas. Smaller spot beams

CLEAN ROOM FACILITY AT ISITE



Spacecraft on Physical Parameters Machine (Mass, C.G, M.I, P.I measurements all in one machine)



ISRO Spacecraft Integration Test Establishment (ISITE), Bengaluru
(Credit: www.isro.gov.in)

increase antenna size. For addressing the wideband needs of satellite testing, PXA and UXA signal analysers and oscilloscopes can be used.

During testing, the satellite's electromagnetic emission capability needs to be measured. A planar near-field scanner is also required for measuring the performance of antennas used for communication.

Challenges in establishing satellite test centres among different nations for space reforms

ESTEC Test Center in Noordwijk, the Netherlands, is the largest satellite test

centre in Europe. STFC RAL Space has been awarded £99 million from the UK government's industrial strategy challenge fund to develop National Satellite Test Facility (NSTF).

At NASA's Goddard Space Flight Centre, a 3.1m × 4.9m robot is busy testing satellites on Earth before it is put to use in space. A six-legged hexapod robot was built by Mikrolar, a New Hampshire-based company.

University of Illinois' premiere space design and testing workspace laboratory supports students, faculty, and other customers to utilise small satellite resources to design, develop, and test satellites. The laboratory is equipped with a clean room, thermal vacuum chamber, vacuum oven, sun simulator, earth orbital magnetic simulator, and range of mechanical and electrical diagnostic equipment. Multiple forms of rapid prototyping design and simulation facilities are also provided.

Satellite Integration and Testing Establishment (SITE) of ISRO was established in 2006. It can integrate and test six satellites of INSAT class at different stages simultaneously. The facility has also carried out vibration and acoustic tests of Mars Orbiter Mission (MOM) spacecraft.

Moving forward

Technological evolution is immensely supporting the satellite testing sector equally. Emerging technologies such as robotics, artificial intelligence, etc, are becoming the fastest drivers of growth and improvements in satellite testing. Opportunities are emerging in the design, manufacturing, and testing of satellites. India has completed a successful decade in satellite testing, and in the coming years, more growth and advancements can be seen in the country's space world. **EFY**



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(Credit: www.esa.int)